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EDUCATION

Indian Institute of Technology (IIT) Dhanbad *Bachelor of Technology in Mining Engineering*

Oct 2021 – May 2025
Dhanbad, Jharkhand

- **Relevant Coursework:** Machine Learning (Supervised & Unsupervised), Deep Learning, Natural Language Processing, Artificial Intelligence, Probability & Statistics, Data Visualization, Python Programming.

EXPERIENCE

Invisible Technologies

Mar 2026 – Present

Python Coding Specialist

Remote – India

- Evaluate **LLM-generated outputs** through **RLHF-style human feedback workflows** — ranking model responses, correcting model outputs and providing structured **preference signals** used to train and improve model accuracy and alignment.
- Design **prompts, evaluation rubrics and expert reference solutions** using advanced **prompt engineering**, used in supervised **fine-tuning** and LLM evaluation pipelines.
- Write, review and debug **Python scripts** generated by LLMs — identifying **logical, syntactic and performance issues** and producing corrected, production-quality code as high-quality training data.
- Perform **rubric-based quality assessments** of model reasoning (chain-of-thought), factual correctness and instruction-following, delivering precise feedback that directly improves downstream **model performance**.

Outlier AI

Nov 2024 – Nov 2025

Coding Expert (Python)

Remote – India

- Designed and authored high-quality **prompts** across coding and analytical domains to elicit and stress-test **LLM responses** as part of **RLHF (Reinforcement Learning from Human Feedback)** training pipelines.
- Evaluated and **compared multiple model-generated responses**, selecting the better output and providing structured **preference rankings, corrections and written feedback** used to fine-tune and improve **model accuracy, reasoning and reliability**.
- Wrote, reviewed, and cleaned **Python scripts** — debugging **LLM-generated code** for logical, syntactic and performance issues and producing corrected, production-quality solutions as model training data.
- Performed **chain-of-thought analysis and failure-mode identification** on model outputs, generating precise training signals that directly contributed to improvements in **response quality and consistency**.

EXTION

Feb 2025 – May 2025

Machine Learning Intern

Remote – Uttar Pradesh, India

- Designed and implemented a production-ready **NLP pipeline in Python** using **NLTK, VADER, Hugging Face Transformers** and **PyTorch** to classify **financial news headlines** into sentiment classes, achieving **92%+ accuracy**.
- Performed extensive **text preprocessing and feature engineering**, including **tokenization, stopword removal and lemmatization**, to improve model generalization and robustness across noisy real-world data.
- Analyzed sentiment outputs at scale and built **visual analytics dashboards** using **Matplotlib** and **Seaborn** to uncover **market sentiment trends** and volatility signals.
- Optimized **GPU-accelerated inference pipelines** using **CUDA** and **cuDNN**, reducing **end-to-end latency by 40%** and enabling near **real-time processing of 10k+ headlines**.

PROJECTS

Smart GenAI PDF Chatbot | RAG, FastAPI, LangChain, FAISS, Docker, GPT-3.5

🐙 GitHub

- Extracted multi-page PDF text using **pdfplumber**, applied **spaCy** sentence tokenization for clean chunking, and generated dense vector embeddings via **all-MiniLM-L6-v2** (SentenceTransformers) indexed into a **FAISS IndexFlatL2** store.
- Built a hybrid query engine with tunable L2 distance threshold — high-confidence matches serve answers directly from PDF; low-confidence queries route through **LangChain PromptTemplate + LLMChain** to **GPT-3.5-turbo** fallback, achieving 100% routing accuracy.
- Exposed the pipeline via a production **FastAPI** backend with **/upload**, **/ask**, **/stats**, and **/health** endpoints, per-request **JSON** monitoring (**query_logs.json**), latency tracking (~23 ms avg), and A/B prompt testing framework.
- Containerized the full stack with **Docker** for one-command reproducible deployment (**docker build + run**), ensuring environment-consistent execution across local and cloud targets.

Bhagavad Gita NLP Chatbot | Semantic Search, Multilingual NLP, FAISS, Gradio

 [GitHub](#)

- Built a multilingual semantic-search chatbot supporting Hindi, English, and Sanskrit using Sentence-BERT embeddings.
- Indexed and queried 700+ Bhagavad Gita verses using FAISS for concept-based, context-aware retrieval.
- Deployed on Hugging Face Spaces with Gradio for seamless real-time interaction without paid APIs.
- Achieved 90%+ answer relevance on philosophical Q&A benchmarks; earned a Kaggle Bronze Medal.

NTT Stock Price Prediction | Machine Learning, LSTM, Time-Series Forecasting

 [GitHub](#)

- Built an end-to-end time-series forecasting pipeline with robust preprocessing, feature engineering, and systematic model tuning.
- Trained and evaluated 8+ ML models (Linear Regression, Random Forest, CatBoost, etc.) achieving up to 95.46% R^2 .
- Developed an LSTM neural network using TensorFlow (Keras) to capture long-term temporal dependencies.
- Forecasted a 30-day stock price trend (\$178.97 $\rightarrow$ \$102.04) with RMSE of 3.09.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: Regression, Classification, Clustering, Model Tuning & Validation, Feature Engineering, Time-Series Forecasting, Deep Learning (CNN, LSTM)

Generative AI & NLP: LLMs, Prompt Engineering, RLHF & Human Feedback Workflows, Fine-Tuning, LLM Evaluation, RAG (Retrieval-Augmented Generation), Semantic Search, Vector Databases (FAISS), Embeddings (SBERT), Transformers, Text Classification, Sentiment Analysis

Frameworks/Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, Sentence-Transformers, LangChain, FastAPI, spaCy, pdfplumber, Matplotlib, Seaborn

Deployment & Tools: Git/GitHub, Docker, GitHub Actions (CI/CD), Streamlit, Gradio, Hugging Face Spaces, AWS (Basic), Azure App Service

ACHIEVEMENTS

- **Kaggle:** 3 $\times$ Bronze Medals in NLP Chatbot, E-Retail Segmentation, and CNN Image Classification.
- **HackerRank:** Earned a **5 rating in Python** by solving multiple coding challenges.
- **HackerRank:** Earned a **5 rating in SQL** by solving advanced database queries and problems.

LEADERSHIP / EXTRACURRICULAR

- Organizer – Khanan'22 Geo-Mining Fest, IIT Dhanbad
- Completed **BCG X Data Science Virtual Experience Program (Forage)**
- Volunteered with **Kartavya NGO** (teaching underprivileged children)
- Cricket enthusiast